

Weekly Report 4

Automated Biometric System for Individual Re-Identification of Mugger Crocodile

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1 Introduction

The goal of this project is to develop an automated biometric system capable of recognizing individual Mugger crocodiles using aerial images. The objective is to determine whether a crocodile observed in a new image belongs to a previously known individual or represents a new individual.

Following the mid-semester evaluation, we refined our approach on how to enhance the quality of the annotations and measure the accuracy of the detection and identification pipeline with fine-tuned models. The major improvements include:

- Reducing the annotation strategy to be specific to the dorsal scutes area.
- Comparing the YOLO detection model and Inception-v3 model based RE-ID model with the refined annotations.

2 Annotation Refinement

One of the problems that were found to be a major problem during evaluation was the annotation strategy. Originally, the bounding boxes were placed on the whole body of crocodiles. However, the methodology produced a lot of background noise like water, mud and vegetation and this adversely affected feature extraction.

To address this, the annotation strategy was refined. Bounding boxes are now drawn only on the area of the dorsal scute having distinct scale patterns which serves as a natural biometric identifier of every crocodile. This change will make sure that there is attention to discriminative identity features, less background noise and better consistency across samples.

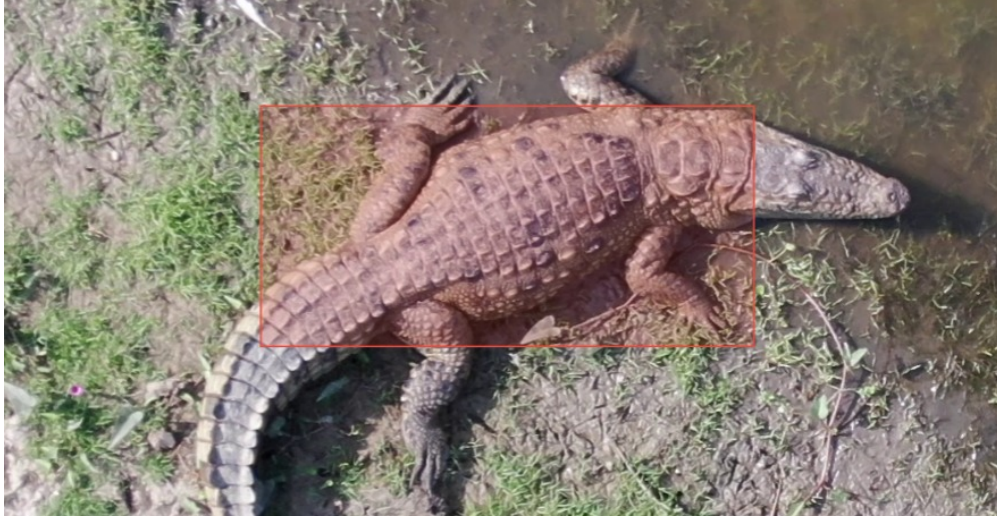


Figure 1: Annotation focused on dorsal scute region for biometric identification

This modification ensures:

- Focus on discriminative identity features
- Reduction of background noise
- Improved consistency across samples

3 Model Implementation

The new pipeline has the following steps:

- **Detection phase:** YOLOv5 is trained to identify the dorsal scute regions.
- **Cropping:** Extraction of the scute regions in all images.
- **Re-identification:** Training of a deep learning model on the detected scute regions.
- **Open-set decision:** Use of a threshold-based classification model to identify known and unknown persons.

4 Expected Improvements

The refined annotation and pipeline are expected to improve system performance in the following ways:

- Improved feature extraction due to focus on dorsal scute patterns
- Better generalization across different time periods
- Reduction in false positives caused by background noise

5 Results (YOLOv5)

The YOLOv5 model was evaluated under the open-set protocol across all test sets.

Table 1: YOLO Predictions on Test-III Dataset

Image	True ID	Predicted	Conf	Boxes	Result
145_first_145_1.jpg	145	109	0.9369	1	False Pos
145_last_145_712.jpg	145	109	0.9550	2	False Pos
146_first_146_1.jpg	146	unknown	0.3146	1	Rejected
146_last_146_651.jpg	146	109	0.9226	2	False Pos
147_first_147_1.jpg	147	unknown	0.7110	1	Rejected
147_last_147_777.jpg	147	unknown	0.7127	2	Rejected
148_first_148_1.jpg	148	unknown	0.8190	1	Rejected
148_last_148_711.jpg	148	109	0.9570	1	False Pos
149_first_149_1.jpg	149	unknown	0.0000	0	Rejected
149_last_149_667.jpg	149	109	0.9505	2	False Pos
150_first_150_1.jpg	150	109	0.8785	2	False Pos
150_last_150_745.jpg	150	unknown	0.4984	1	Rejected
151_first_151_1.jpg	151	109	0.9494	1	False Pos
151_last_151_737.jpg	151	109	0.8876	2	False Pos
152_first_152_1.jpg	152	109	0.8611	1	False Pos
152_last_152_673.jpg	152	unknown	0.7058	1	Rejected
153_first_153_1.jpg	153	109	0.9200	1	False Pos
153_last_153_634.jpg	153	unknown	0.8237	1	Rejected
154_first_154_1.jpg	154	109	0.9544	1	False Pos
154_last_154_736.jpg	154	109	0.9558	1	False Pos
155_first_155_1.jpg	155	unknown	0.8334	1	Rejected
155_last_155_719.jpg	155	unknown	0.5210	2	Rejected
156_first_156_1.jpg	156	109	0.8806	1	False Pos
156_last_156_751.jpg	156	109	0.8782	1	False Pos
157_first_157_1.jpg	157	109	0.9594	2	False Pos
157_last_157_733.jpg	157	109	0.9372	3	False Pos
158_first_158_1.jpg	158	unknown	0.8261	3	Rejected
158_last_158_693.jpg	158	109	0.9162	3	False Pos
159_first_159_1.jpg	159	109	0.9497	1	False Pos
159_last_159_607.jpg	159	109	0.9206	2	False Pos
160_first_160_1.jpg	160	unknown	0.7423	1	Rejected
160_last_160_711.jpg	160	unknown	0.4371	1	Rejected

5.1 Observations

- **Test-I:** Perfect re-identification (TPR: 100%) with moderate unknown rejection (TNR: 85.09%).
- **Test-II:** Acceptable unknown rejection (TNR: 88.75%), but presence of false positives.
- **Test-III:** Poor generalization with low TNR (40.62%) and high false positives (59.4%).
- **False Positives:** Many unknown individuals incorrectly classified as known identities, frequently biased toward a single class.

6 Challenges Faced

The following challenges were encountered:

- Background noise in complete images is an issue that troubles the model learning process.
- Problems with differentiating between familiar and unfamiliar crocodiles (open-set problem)
- Retrospective attention towards dorsal scute area in previous annotations.
- The changes in UAV images including light and viewing angles.

7 Plan for Next Week

According to the constraints noted with the current configuration and experiences of the reference pipeline, the second step will be the transformation of single model evaluation into a formal re-identification pipeline.

The steps to be planned are as follows:

- Trained YOLO model is used to localize dorsal scute regions in the entire images.
- Use to extract cropped dorsal scute areas of all images and save them. These cropped areas can be fed into the Inception-v3 model as opposed to full images.

This move is inspired by the fact that the presence of background noise affects performance on classification adversely, and learning features on the scute area is enhanced.